

Wärtsilä 25 Ammonia

MISC

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Wärtsilä 25

High & green performance

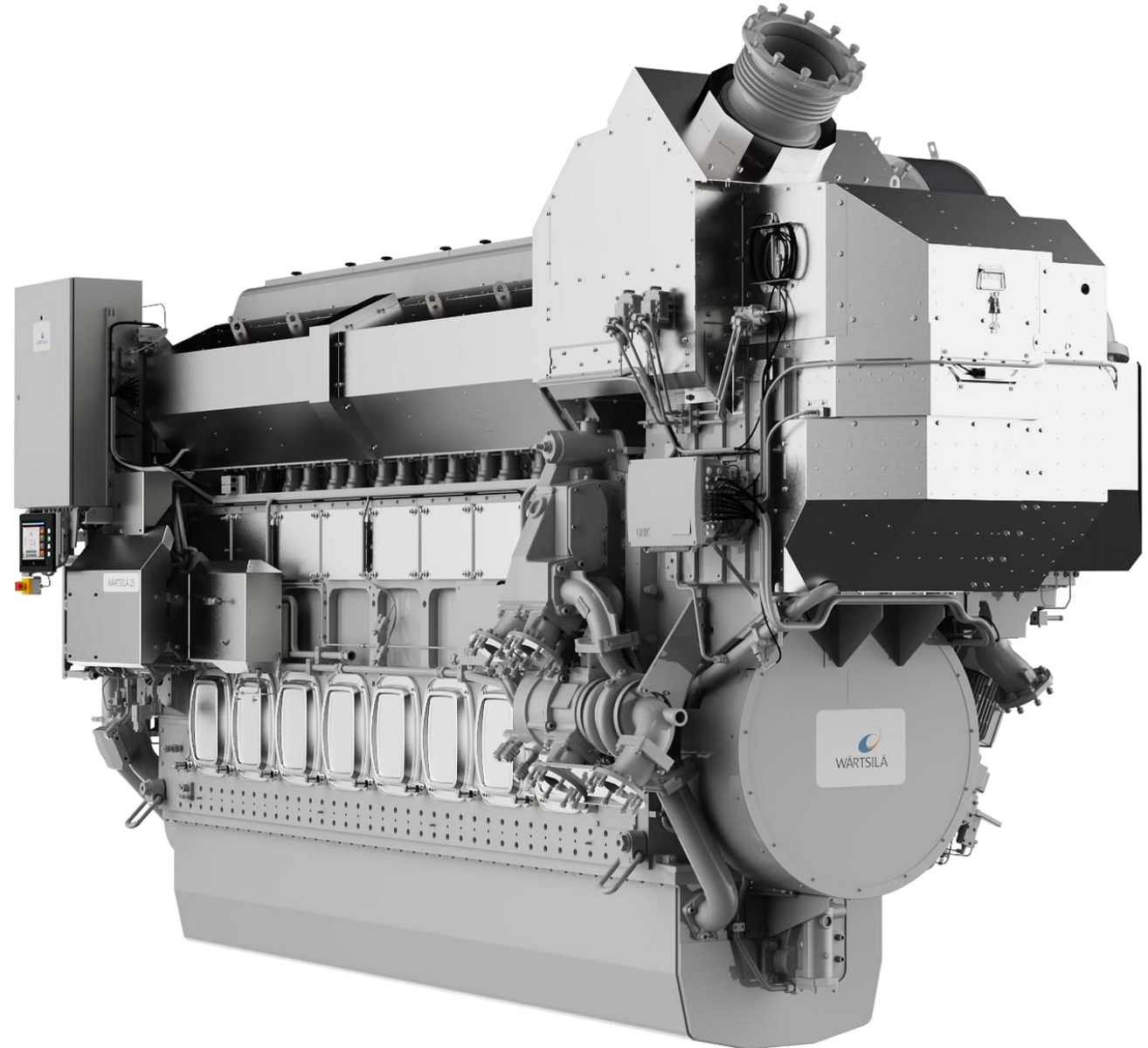
New Dual Fuel engine for 1.6-3.4
MW power range

Cost-efficient & reliable

Longer time between overhauls
Easy maintenance with common
components

Future-proof & safe

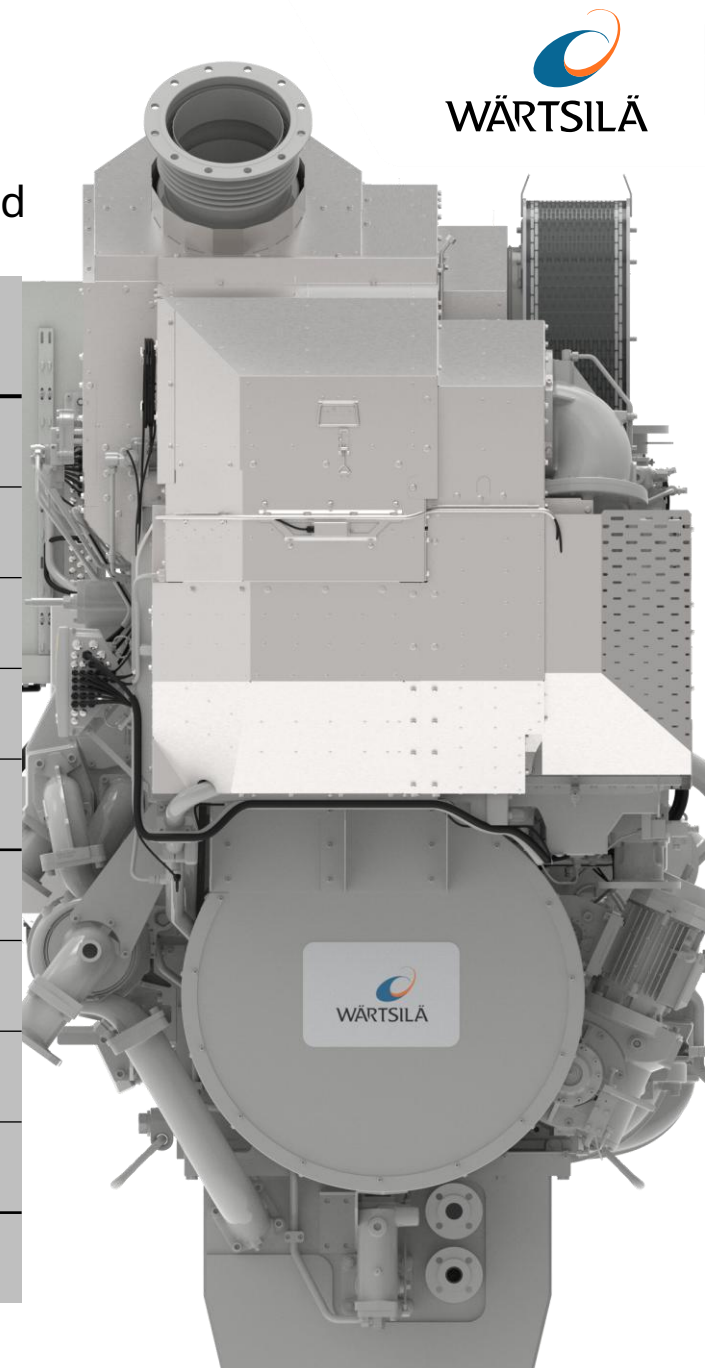
Modular structure offers
flexibility towards future fuels
and secures your investment
in targeting net-zero



Engine family overview

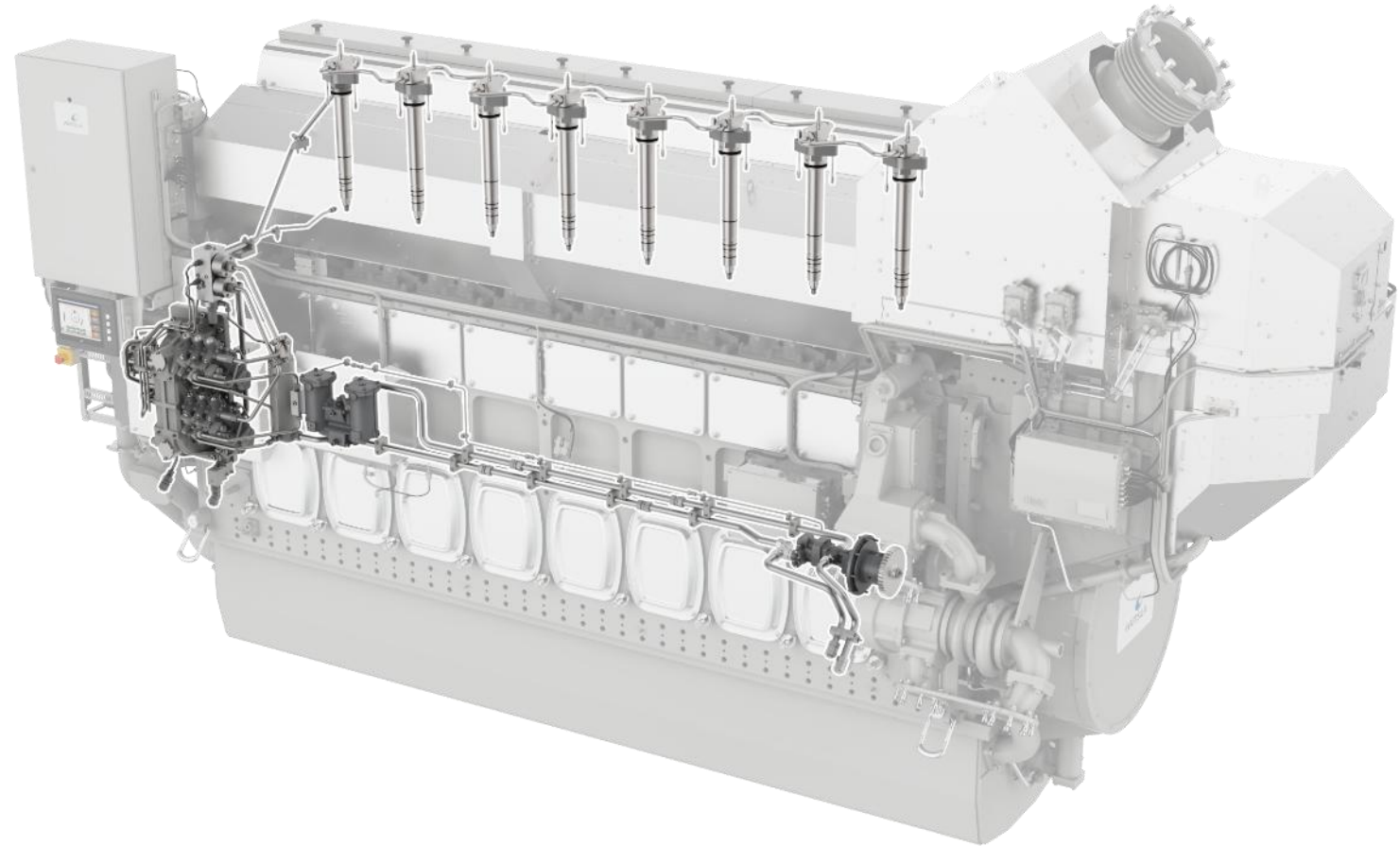
Flexible engine for high and green performance. Suppliers from Europe and manufactured in Finland.

	Wärtsilä 25		Wärtsilä 25DF		Wärtsilä 25DF Ammonia	
Cylinder bore (mm)	250		250		250	
Piston stroke (mm)	340		340		340	
Nom speed (rpm)	900	1 000	900	1 000	900	1 000
Power / cyl. (kW)	345	375	315	345	315	345
BMEP (MPa)	2.72	2.70	2.52	2.48	2.52	2.48
6L power (kWm)	2 070	2 250	1 890	2 070	1 890	2 070
7L power (kWm)	2 415	2 625	2 205	2 415	2 205	2 415
8L power (kWm)	2 760	3 000	2 520	2 760	2 520	2 760
9L power (kWm)	3 105	3 375	2 835	3 105	2 835	3105
Application	DM, DE, AUX		DE, AUX	DM, DE, AUX	DE, AUX	



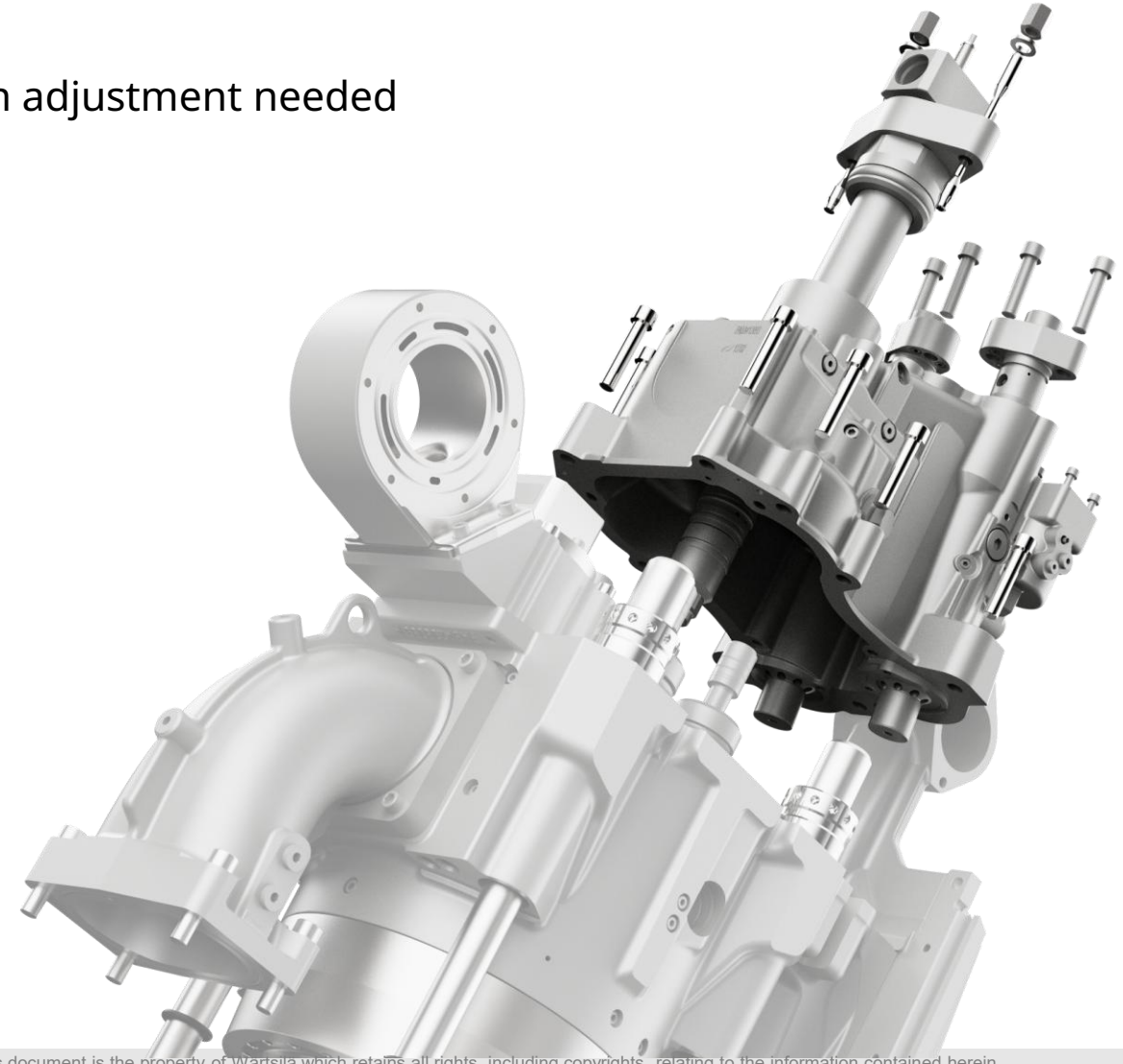
Common rail fuel injection system for accurate and fast control

- Fully electronic system with ultra-precise injection capability
- Independent operation cylinder per cylinder
- Same circuit for pilot and main fuel
- Injectors with single needle technology, same needle for diesel and pilot injection in DF operations.
- Smokeless operation in all conditions
- Low emissions
- High efficiency at part load
- Cylinder-wise "trip to diesel" capability



Superior combustion control with variable valve-timing

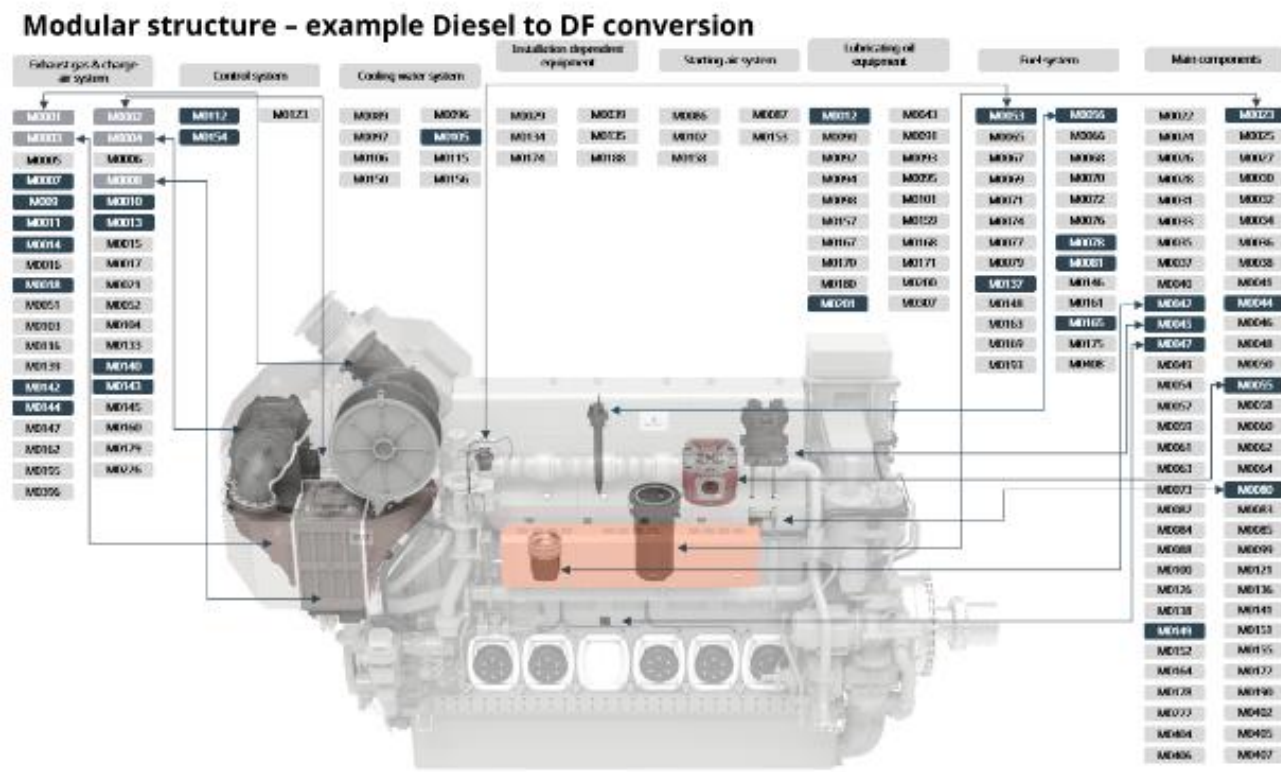
- Both on-off type and stepless available
- Hydraulic valve mechanism - better accuracy and no valve lash adjustment needed
- Enabler for higher efficiency in the whole load range
- Low emission optimization
- Optimal valve timing at each load
- Reduced smoke and particulate matter emissions
- Valve timing flexibility is also an enabler for future fuels



Focus on product modularity

- High consistent quality through less component variants
- High commonality between engine version for efficient assembly and streamlined procurement process.
- Standardized maintenance procedures, supporting low-manning maintenance concepts
- Facilitate future product upgrades and conversions

Design for future.



Modular service concept supports the efficiency and flexibility of Wärtsilä 25

- Engine is designed for long periods of maintenance-free operation and has a TBO of up to 32,000 hours (24'000 hours with ammonia)
- From single spare parts to exchange units:
 - Cylinder heads , injectors and high-pressure fuel pumps
- Easier planning through work cards and more use of service kits & sets
- Power units:
 - the pre-overhauled cylinder unit can be exchanged quickly
 - also the traditional, piece by piece, maintenance can be conducted
- Data-driven dynamic maintenance planning with remote operational support
- Fuel conversion Diesel ↔ DF
 - the modular design enables fast and efficient conversion
 - standardized component interfaces implies no need for machining

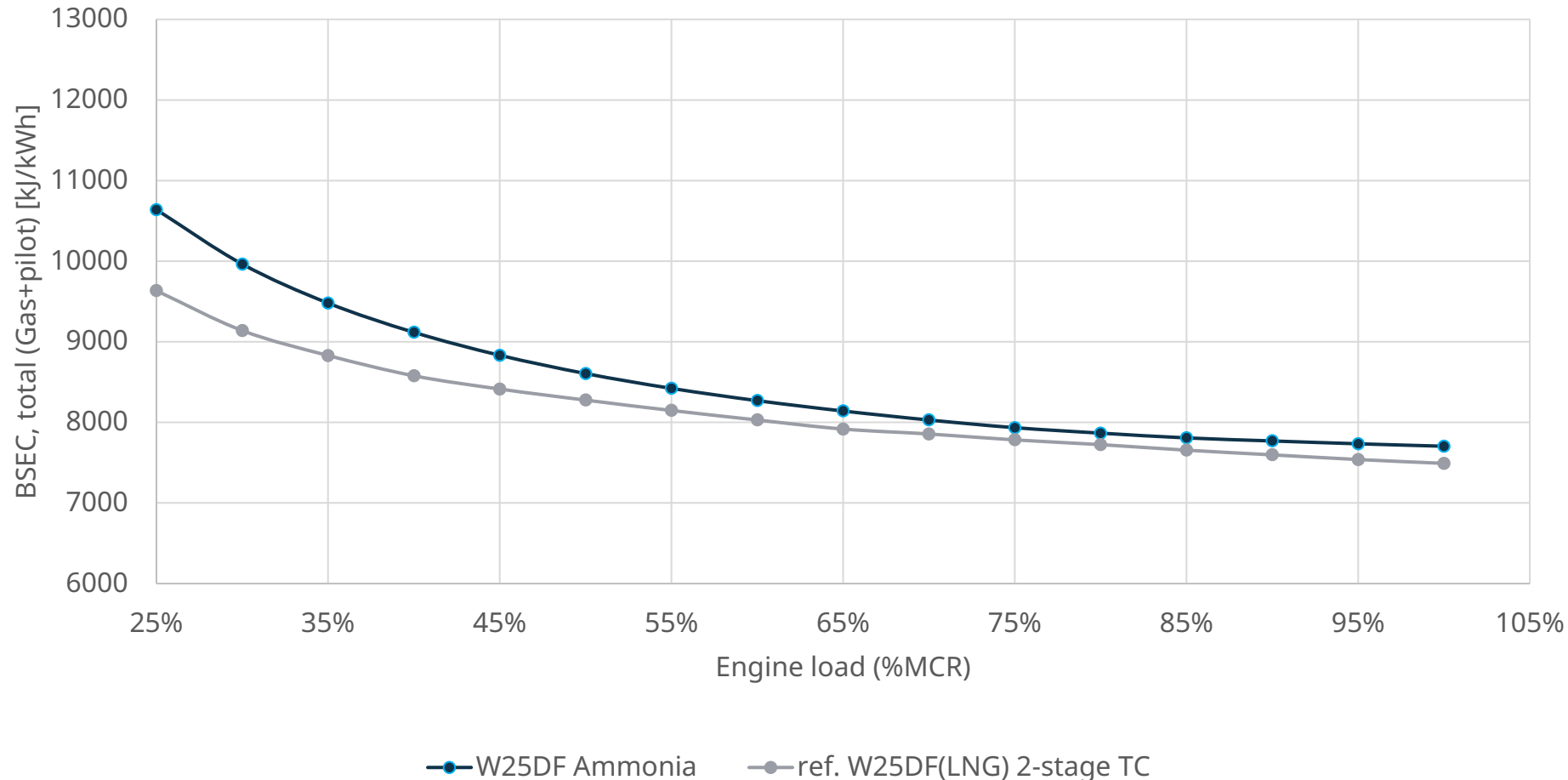


W25 Ammonia testing has continued to our full satisfaction

- Since the launch in November 2023, the ammonia solution has undergone extensive testing to optimize its performance
- Reduction in total GHG emissions now above 90% (Tank to Wake) [press release 27.5.2025](#)
- Power increased from 280/305 to 315/345 kW/cyl, same as W25DF output.
- Energy share of ammonia have raised from ~85% to ~ 95%
- Engine efficiency increased to 47%
- EIAPP and Type Approval successfully performed
- 4 vessels with W25 ammonia engine contracted (1.4.2026)



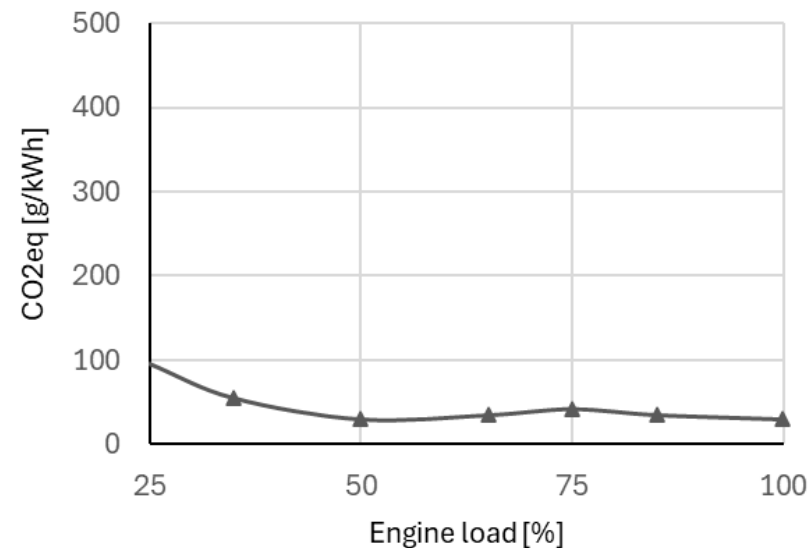
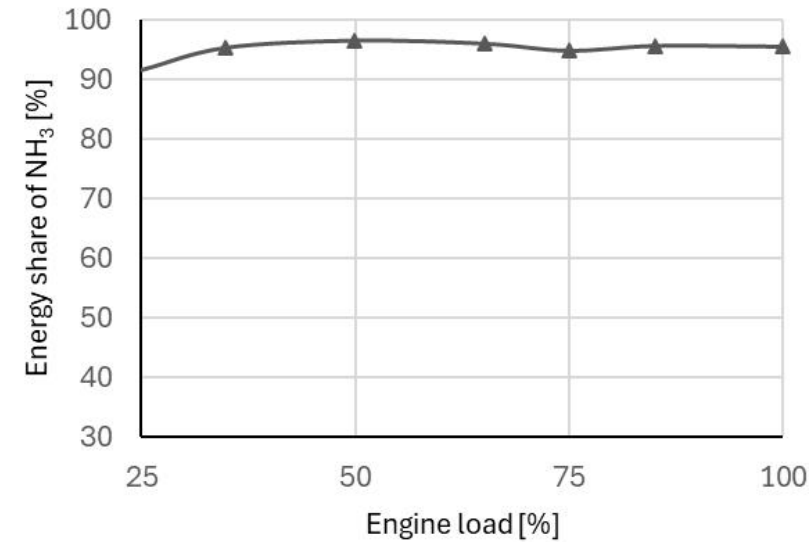
Energy consumption as of today correspond to an engine efficiency of 46,7%,



Fuel consumption according to ISO 15550. LFO lower calorific value 42 700 kJ/kg including engine driven pumps, tolerance 5%. IMO Tier III, 70mbar backpressure allowance in NH₃.

Wärtsilä 25 Ammonia share

- Ammonia is ignited by a small amount of pilot fuel.
 - The energy share of ammonia is ~95% across a wide load range.
 - Fuel efficiency of the low-pressure Otto concept has been consistent with and equal to that of dual fuel LNG engines.
-
- CO₂ equivalent emissions are reduced over 90% (TtW) compared to a diesel engine using LFO.
 - The CO₂ equivalent emissions originate from the pilot fuel used and the N₂O formed during combustion.
 - Ammonia has proven to be a good fuel for the low-pressure Otto concept.





System overview

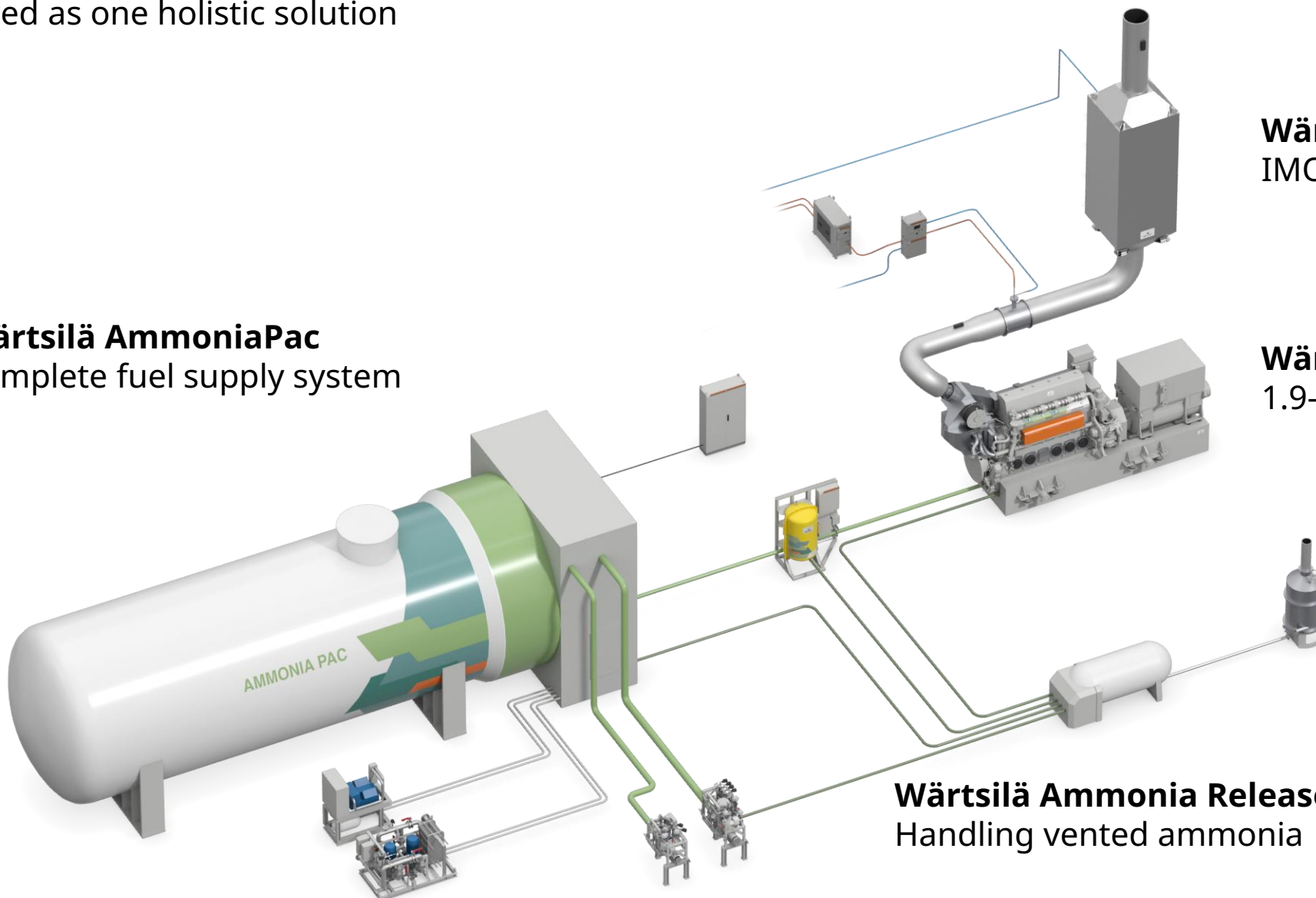
Engineered as one holistic solution

Wärtsilä AmmoniaPac
Complete fuel supply system

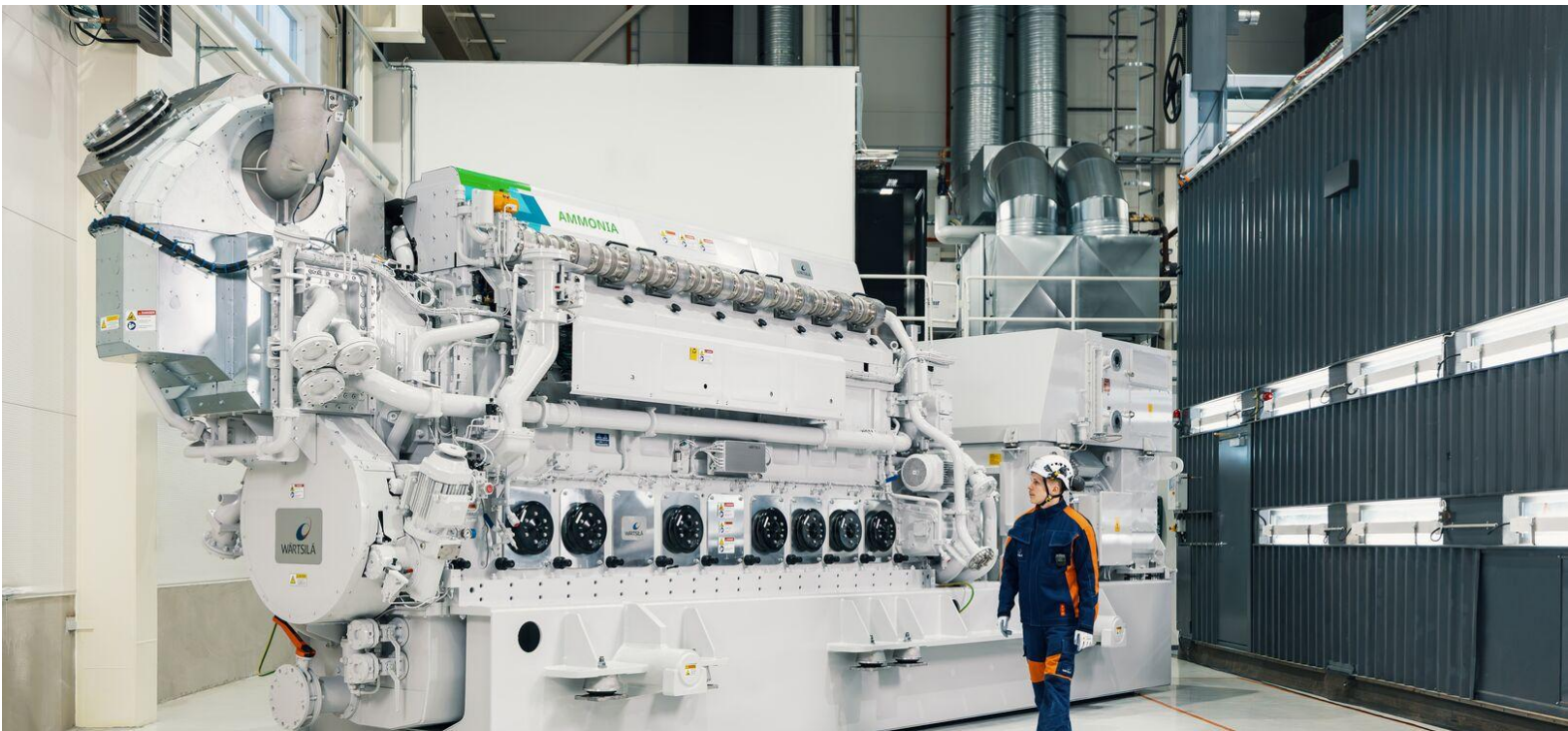
Wärtsilä NOx Reducer
IMO Tier II / Tier III

Wärtsilä 25 Ammonia
1.9-3.1MW

Wärtsilä Ammonia Release Mitigation System
Handling vented ammonia



From concept to hardware



Wärtsilä 25 Ammonia engine underwent finishing work at WaskiCenter, Vaasa.



AmmoniaPac fuel tank ready for transportation

Gas valve unit (GVU)



ARMS buffer tank

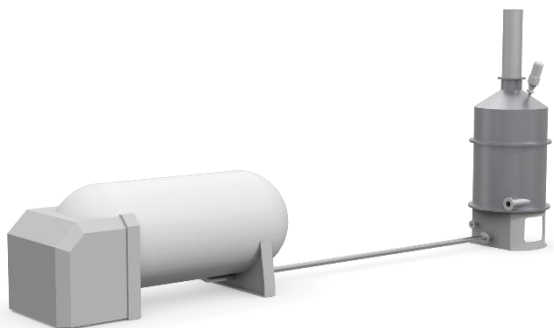


The AmmoniaPac fuel tank, gas valve unit (GVU) and ammonia release mitigation system's (ARMS) buffer tank are essential parts of the Wärtsilä 25 Ammonia solution.

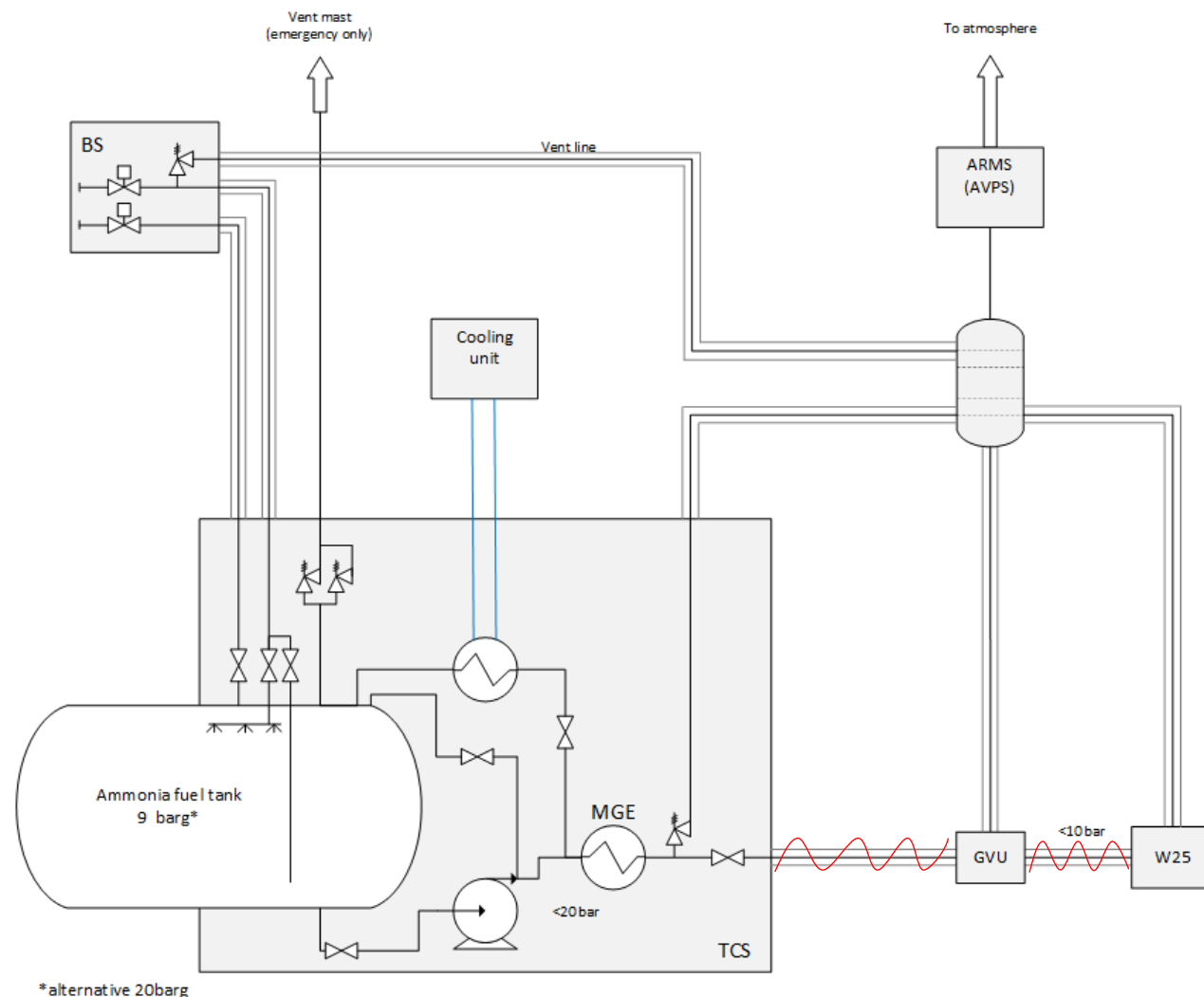
AmmoniaPac – fuel gas supply system (cont.)

Wärtsilä Ammonia Release Mitigation System (WARMS)

- All vented Ammonia from the system is led to the ARMS collection tank (except tank PRVs)
- The collected gas is burned off in the burner, resulting in practically zero emissions
- Uses ammonia as pilot fuel to avoid additional CO₂ emissions



Heat tracing on fuel piping to GUV and engine to keep Ammonia from condensing



PSV conversion for Ammonia for Eidesvik Offshore

- Wärtsilä partners with Norwegian shipowner Eidesvik to supply the equipment for the conversion of the PSV Viking Energy for ammonia operation, contract was signed in June 2024
- The vessel is under contract to energy major Equinor, through the conversion emission will be cut by at least 90 percent
- The conversion project is planned for early 2026, with final commissioning ongoing and vessel start operation end 2026.
- Scope:
 - 1x 9L25 Ammonia
 - AmmoniaPac Fuel Gas Supply System
 - Ammonia Release Mitigation System
 - Wärtsilä NOx Reducer



In October 2025 we signed our second Ammonia project - W25 Ammonia will power a 7800DWT Bulk Carrier for Skarv Shipping II AS built at Huanghai

- Skarv Shipping II AS is owned by Grieg Edge and Peak CSL group, based in Bergen NO
- Shipyard Huanghai Shipbuilding Co. Ltd
- Vessel building start – Q2 /2026
- Vessel delivery expected Q3 /2027
- 1 x W6L25 Ammonia genset as main engine + SCR, delivery Q4/26
- AmmoniaPac with bunker station, 160 m3 pressurized tank, WARMS and GVU, delivery Q1/27



“We see ammonia as one of the future fuels and are happy to see strong European companies like Wärtsilä leading the push towards carbon-free propulsion. The Wärtsilä 25 Ammonia solution promises exactly what we are looking for: robust safety and environmental measures combined with advanced energy efficient technology,”

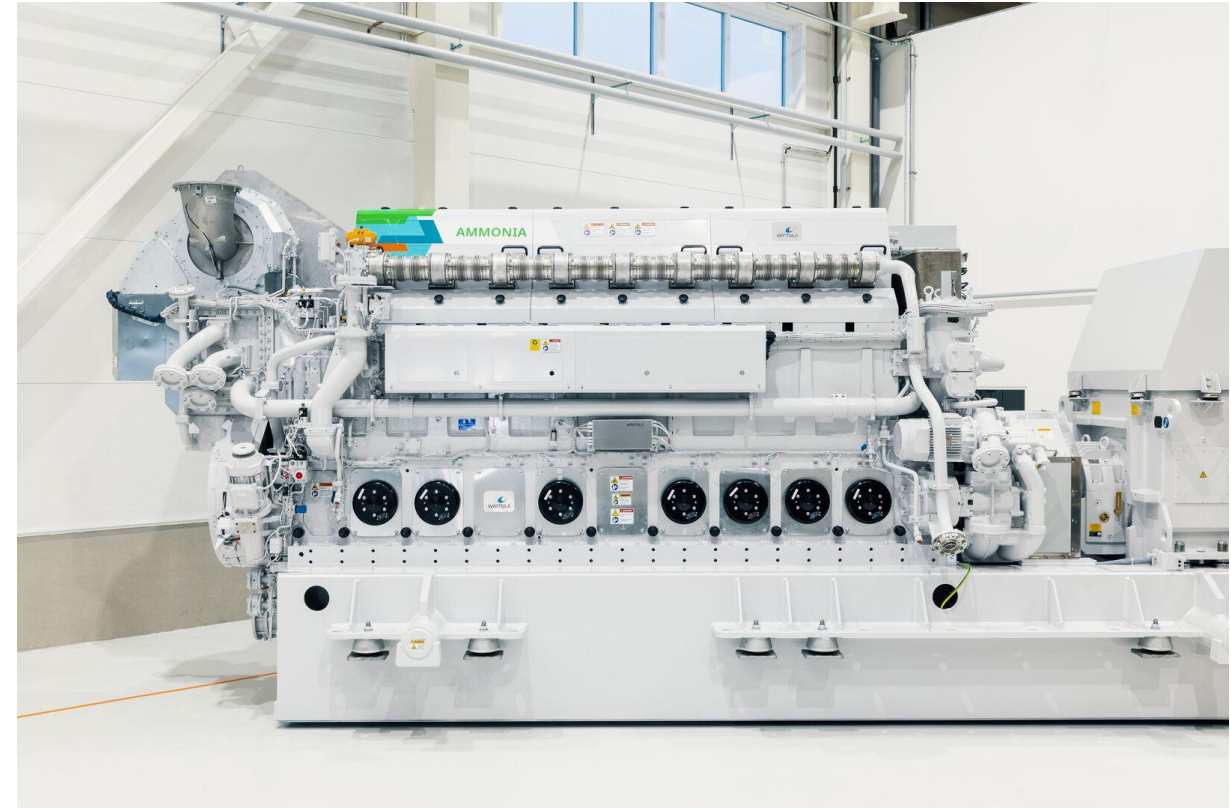
Jan Øivind Svardal, CEO – Skarv Shipping Solutions.

2x 51,350 m³ LPG/liquid ammonia carrier vessels for Navigator Amon Shipping, build at Nantong CIMC Sinopacific Offshore & Engineering in Shanghai

“Wärtsilä has demonstrated industry-leading capabilities in enabling future low and zero-carbon fuels to become viable. By selecting the Wärtsilä 25 Ammonia engine, we expect to dramatically reduce the ships’ environmental impact, while supporting compliance with evolving regulatory requirements,” says Mr. Kaj S. Pilemand, Director of Vessel Operations, Navigator Gas.

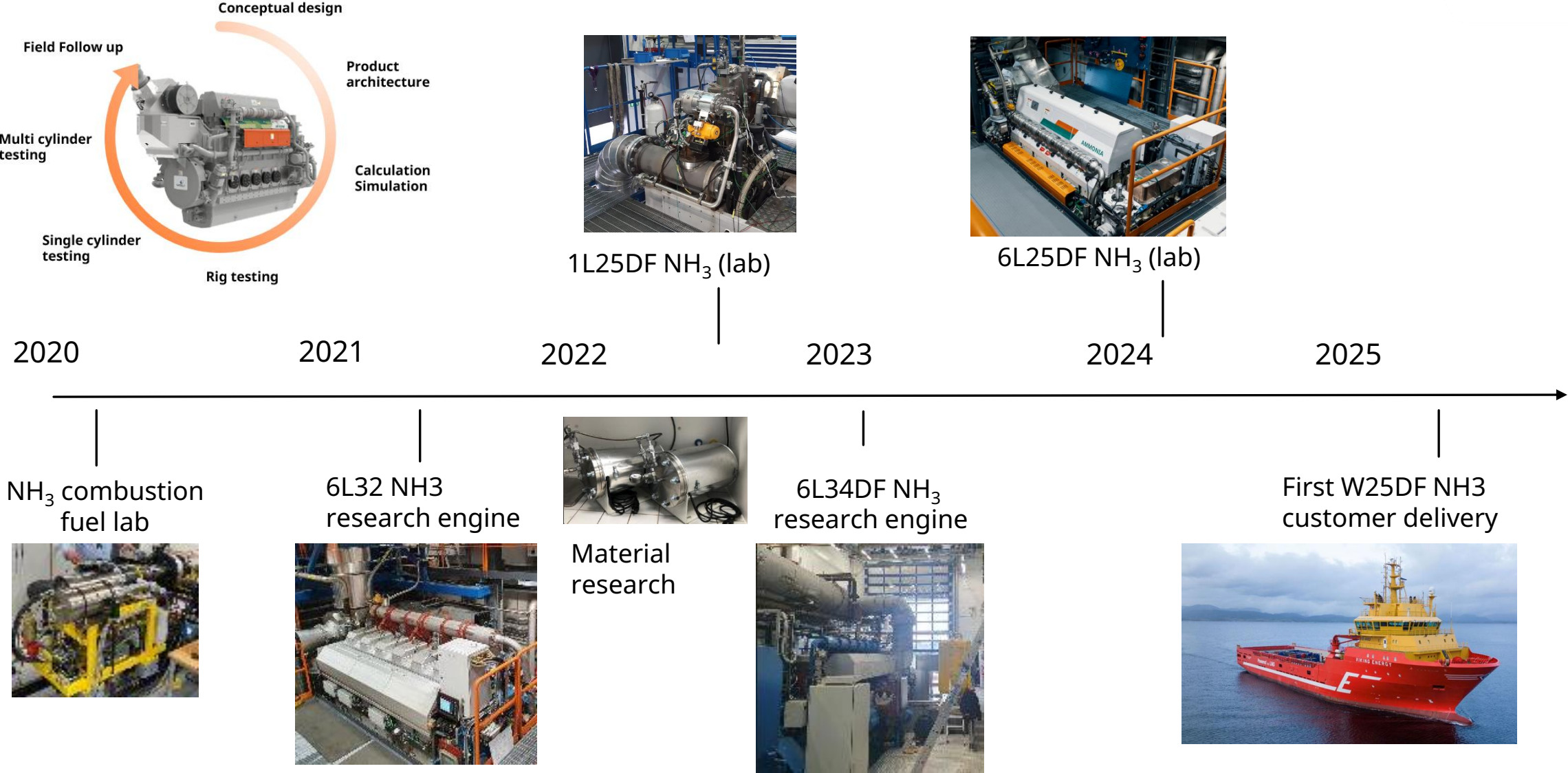
Two new gas carriers to stay ahead of environmental standards with Wärtsilä 25 Ammonia engine

- Wärtsilä scope of supply includes Wärtsilä 8L25 Ammonia auxiliary engine, GVU, SCR & Controllable Pitch Propulsion solution.
- Engines delivery for first ship end of May 2027 and second ship End of September 2027.





From research to commercial product



Testing in laboratory on multiple full-scale engines

- No leakage or smell of ammonia during engine operation or after the engine is stopped.
- Control of ammonia gas temperature and gas piping temperature is important to avoid condensation.
- After running on ammonia, engine is run in diesel mode before stopped.
- When it comes to maintenance, the primary distinction between conventional fuels and ammonia lies in the preparatory work conducted before initiating a work.
- Ventilation & gas detection.



Wärtsilä 6L34DF ammonia research engine at Sustainable Energy Katapult center testing facility in Norway.

Why did we go for Otto concept on Wärtsilä 25 platform?

- Wärtsilä have tested both Otto and Diesel concept during the development of Ammonia engine.
- Conclusion of those tests were that Wärtsilä selected Otto (low pressure) concept due to:
 - **Safety point-of-view.** With low pressure concept you have much lower operating pressures → less complex compared to HP
 - **Performance/efficiency.** With Diesel (high pressure) concept you need to have high pressure pump to get the pressure high enough before injecting to cylinders, that will take approx 3%-unit of the engine efficiency.
 - **Emissions (NO_x, N₂O, NH₃):** SCR is required for both, low pressure and high-pressure concept to be able to reach IMO Tier 3 level. N₂O slip on same level with low pressure concept. NH₃ slip after SCR on similar level with both concepts.
 - **Loading performance.** During testing & validation phase we have improved loading performance. We see that the loading performance will be very similar to a LNG DF engine or even better in the future once optimization is done.
 - **Knocking/misfiring.** With low pressure ammonia concept we have seen that engine is even less sensitive to misfiring and knocking compared to LNG DF.

Safe and reliable operations ensured

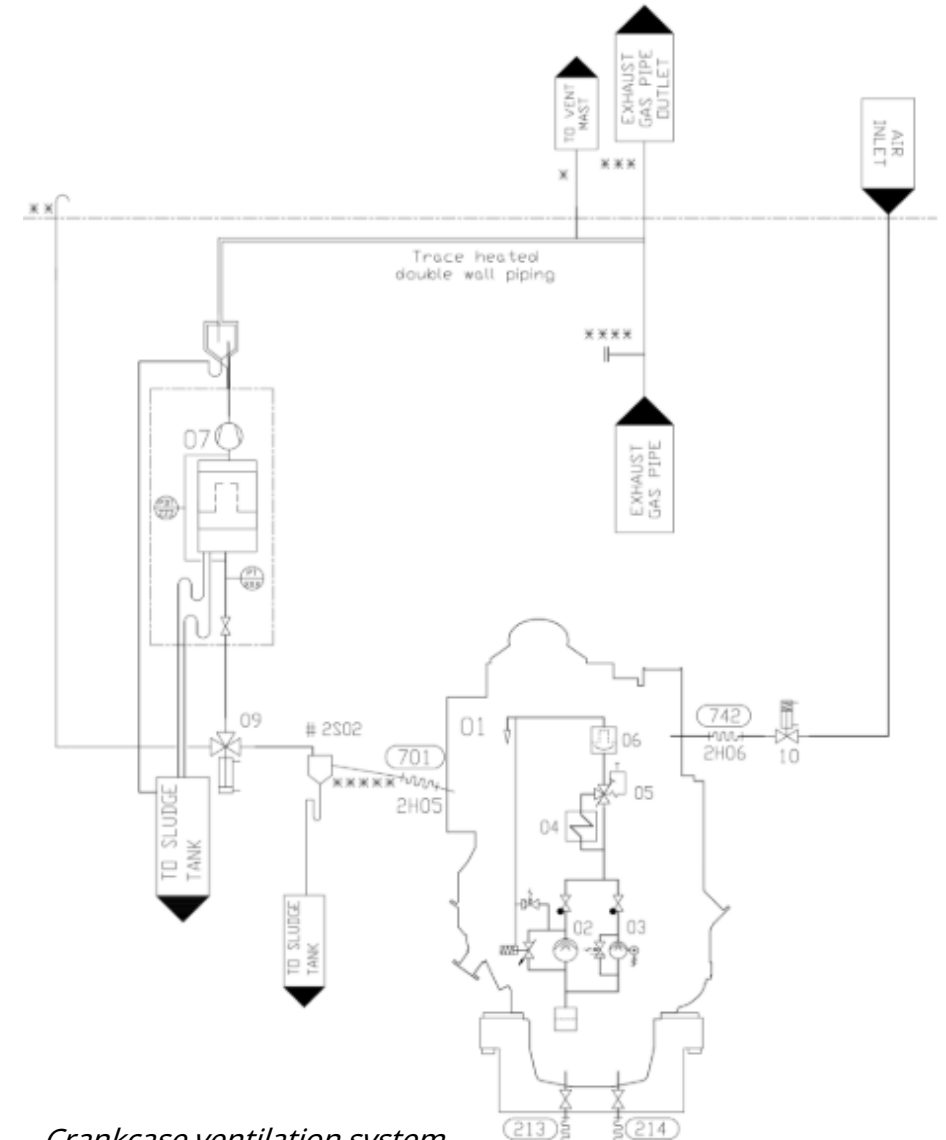
- Foundation for the ammonia solution is well proven LNG safety principles
- Safety concept developed in collaboration with classification societies (AiP granted by DNV in 2023 and BV in 2024)
- Due to the toxicity of ammonia the main difference to LNG DF are the maintenance procedures and engine auxiliary systems
- PPE to be easily available, including gas detection
- No restrictions for personnel entering engine room in normal operation thanks to gas-safe engine room design principle. Engine start and stop in diesel mode.
- Integrated automation system ensuring safe operations



Wärtsilä 25DF Ammonia – safety considerations & machinery protection

Main differences compared to proven DF (LNG) engine:

1. Gas pipe temperature monitoring added for DF Ammonia – preventing condensation of gaseous ammonia in gas pipe
2. Crankcase pressure, same functionality but lower setpoint for DF Ammonia – suction (negative pressure) in crankcase for engine room safety & smell avoidance
3. Pressure monitoring in engine gas manifold degassing line added for DF Ammonia due to ARMS



Crankcase ventilation system

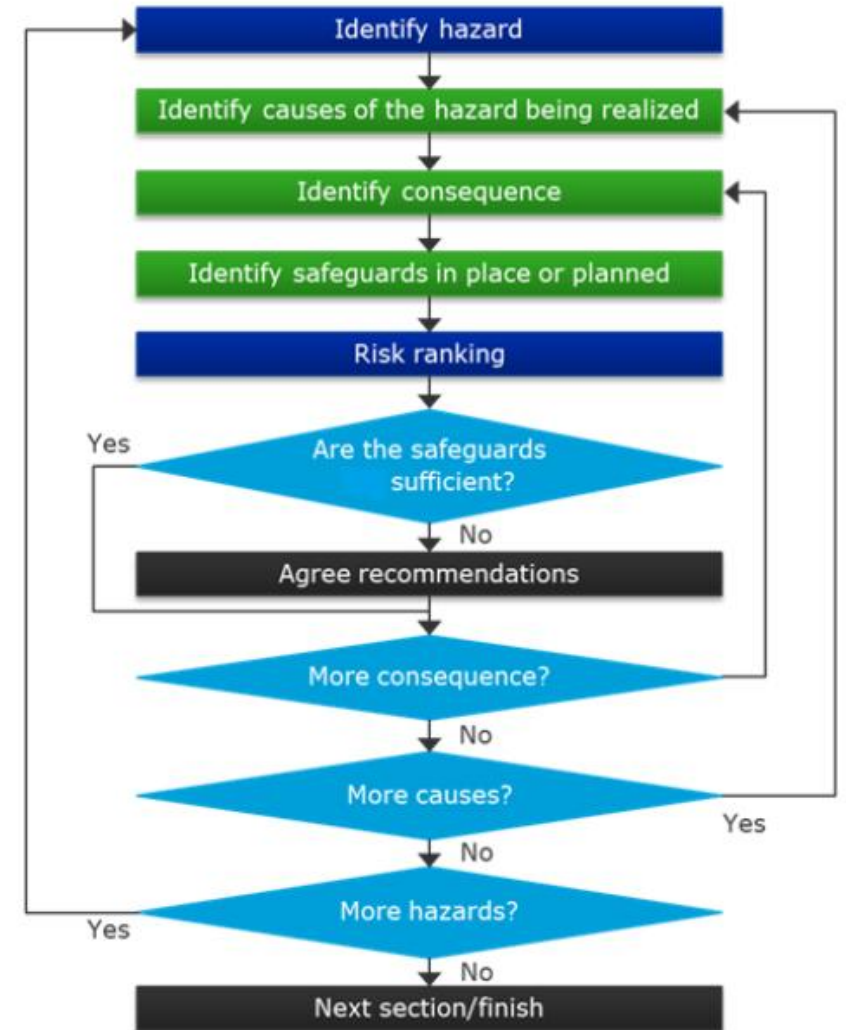
HAZID & HAZOP – procedure and experiences

HAZID

- High-level hazard/risk identification process for the conceptual design following a formal methodology
- Normally chaired by Classification society representative, alternatively by consultant
- Participants may be, but not limited to, for example; ship designer, class society, equipment supplier, consultant, ship owner, shipyard, flag state authority

HAZOP

- Detailed hazard/risk identification process based on a finalised, project-specific system design including cause-and-effect diagram
- Review of failure mechanisms for each and every component and function selected for the system



HAZID methodology visualised by DNV

Key takeaways

- ✓ Total greenhouse gas emissions reduced at minimum 90%
- ✓ Engine efficiency ~ 47%
- ✓ Ammonia energy share ~ 95%
- ✓ Same power with ammonia as LNG DF
- ✓ Dual-fuel technology with full flexibility to run in ammonia mode or diesel mode
- ✓ Complete solution - fuel gas supply system, engine and aftertreatment system
- ✓ Integrated automation system ensuring safe operations
- ✓ Wärtsilä offers training services and a global service network available 24/7





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